

Locally Honest Coverage Forces Nonparametric Inflation: a Sharpness Lower Bound for Prediction Sets

Peter Cotton

`peter.cotton@microprediction.com`

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Abstract

Marginal conformal coverage costs no sharpness and admits oracle-rate bands. A guarantee at a fixed test point is different. For any set with honest coverage at an interior point, valid over a Hölder class of residual laws, we show the realized threshold exceeds the oracle by the local modulus $(ng(x_0))^{-s/(2s+d)}$ on a majority of samples. The cause is Le Cam indistinguishability, a testing-power floor, not exchangeability. Minimax-rate sharpness is still compatible with honesty. Zero excess and adaptation to unknown smoothness are not. A plug-in estimate is sharper because it declines to insure against alternatives the data cannot rule out.

1 Two guarantees, one honest

Fix a model, freeze it, and let $R = A(X, Y)$ be its scalar nonconformity score with conditional law $\mathcal{L}(R | X = x)$. Ordinary conformal coverage is

$$\mathbb{P}\{R_{n+1} \in C_n(X_{n+1})\} \geq 1 - \alpha, \quad (1)$$

the probability averaging over the test point X_{n+1} and the calibration sample D_n . The honesty lower bound below does not follow from marginal validity: Lei and Wasserman (2014) construct bands with exact finite-sample marginal validity that converge to an oracle band at the minimax rate, and localized conformal methods retain marginal validity while weighting by proximity. The cost we identify attaches not to marginal coverage but to a guarantee at a fixed point, formalized next as honest local coverage.

Definition 1 (Honest local coverage). Fix an interior point x_0 and $0 \leq \eta < \frac{1}{2}$. A one-sided prediction set $C_n(x_0) = [0, T_n(x_0)]$ for a nonnegative score is *honestly locally valid* over a class Θ if

$$\inf_{\theta \in \Theta} \mathbb{P}_{\theta} \left\{ F_{\theta(x_0)}(T_n(x_0)) \geq 1 - \alpha \right\} \geq 1 - \eta, \quad (2)$$

where $F_{\theta(x_0)}$, assumed continuous and strictly increasing, is the true conditional law of the score at x_0 , so the event equals $\{T_n(x_0) \geq q_{\alpha}(\theta(x_0))\}$, and the probability is over the calibration sample. In words: with probability at least $1 - \eta$ over the data, the realized set has conditional coverage $1 - \alpha$ at x_0 , whatever law in Θ generated the data. Taking $\eta = 0$ is literal per-sample conditional coverage. This is a training-conditional, fixed- x_0 requirement, the local analogue of

the training-conditional and tolerance-region guarantees studied for conformal sets (Vovk, 2012), and it is deliberately stronger than the marginal coverage (1).¹

Sharpness is measured against the oracle threshold $q_\alpha(u) = F_u^{-1}(1 - \alpha)$ through the excess

$$S_\theta(T_n) = \mathbb{E}_\theta \left[\left(T_n(x_0) - q_\alpha(\theta(x_0)) \right)_+ \right], \quad (3)$$

the expected inflation of the threshold above the oracle. The question is how small (3) can be made subject to (2).

2 Assumptions

The upper Hölder bound alone is not enough for a lower bound; the class must contain alternatives that are indistinguishable near x_0 . Three standing assumptions supply this without asking us to estimate an arbitrary law.

Assumption 1 (Local design regularity). The covariate density g satisfies $0 < g_- \leq g(x) \leq g_+ < \infty$ on a fixed neighbourhood of $x_0 \in (0, 1)^d$, and x_0 is a Lebesgue point of g . The Lebesgue point is used only to write the local mass as $g(x_0) h^d(1 + o(1))$; without it every rate below holds verbatim with $g(x_0)$ replaced by constants between g_- and g_+ .

Assumption 2 (Regular residual family). The calibration sample $(X_i, R_i)_{i=1}^n$ is i.i.d. with $X_i \sim g$ and $R|X = x \sim F_{\theta(x)}$ for a one-parameter family $\{F_u : u \in [\underline{u}, \bar{u}]\}$ that is, locally in u , quadratically regular in information and Lipschitz-regular in quantile:

$$\text{KL}(F_u, F_v) \leq K(u - v)^2, \quad c_q |u - v| \leq |q_\alpha(u) - q_\alpha(v)| \leq C_q |u - v|.$$

A truncated exponential family with parameter in a compact interval is a concrete instance. The lower bound below uses only the quadratic information bound and the lower quantile-sensitivity constant c_q ; the upper constant C_q and the design lower bound g_- of Assumption 1 enter only the attainability statement of Section 4.

Assumption 3 (Hölder smoothness with local richness). The parameter field lies in a Hölder ball, $\theta \in \mathcal{H}^s(L)$. Fix a bump profile ψ with $\psi(0) = 1$, $\text{supp } \psi \subseteq B(0, 1)$, and enough smoothness to lie in $\mathcal{H}^s$. There are $h_0, a_0 > 0$ and an interior constant baseline value $u_0 \in (\underline{u}, \bar{u})$ such that

$$\theta_{h,a}(x) = u_0 + a L h^s \psi\left(\frac{x - x_0}{h}\right) \in \mathcal{H}^s(L), \quad 0 < h < h_0, \quad 0 < a < a_0.$$

Assumption 3 is the operative one. It says the class holds a pair that agree outside a ball of radius h around x_0 but differ by $\delta = a L h^s$ at x_0 , and that the amplitude factor a can be shrunk freely below a_0 , which the proof uses to place the two hypotheses at any chosen separation.

¹*Honest* is the standard term for a coverage guarantee that holds uniformly over the class at finite n , the $\inf_\theta$ sitting inside the probability, as opposed to a pointwise or asymptotic guarantee that fixes θ first (Li, 1989; Low, 1997; Cai and Low, 2004). The qualifier *local* is ours: we impose it at a fixed interior point x_0 rather than over an interval.

3 The lower bound

The bound is at bottom a testing-power statement, so we isolate that step. For laws P_1, P_0 on the calibration sample write the Neyman–Pearson frontier

$$\beta_{1-\eta}(P_1, P_0) = \inf_{A: P_1(A) \geq 1-\eta} P_0(A), \quad (4)$$

the least P_0 -mass of any event whose P_1 -mass is at least $1 - \eta$.

Lemma 2 (Honesty is a test, and a test has a floor). *Let θ_0, θ_1 be two parameter fields with $q_1 := q_\alpha(\theta_1(x_0)) > q_\alpha(\theta_0(x_0)) =: q_0$, and let T_n satisfy honest local coverage (2) under θ_1 . Then*

$$\mathbb{P}_{\theta_0}\{T_n(x_0) \geq q_1\} \geq \beta_{1-\eta}(P_{\theta_1}^n, P_{\theta_0}^n), \quad S_{\theta_0}(T_n) \geq (q_1 - q_0) \beta_{1-\eta}(P_{\theta_1}^n, P_{\theta_0}^n).$$

Proof. Honesty under θ_1 makes $A = \{T_n(x_0) \geq q_1\}$, a function of the calibration sample, satisfy $P_{\theta_1}^n(A) \geq 1 - \eta$, so $P_{\theta_0}^n(A) \geq \beta_{1-\eta}$ by (4). On A the excess $T_n(x_0) - q_0$ is at least $q_1 - q_0$, giving the second inequality. The frontier obeys the Le Cam bound $\beta_{1-\eta}(P_1, P_0) \geq 1 - \eta - \text{TV}(P_1, P_0)$, since any A with $P_1(A) \geq 1 - \eta$ has $P_0(A) \geq P_1(A) - \text{TV}$. $\square$

The frontier (4) is exact; the $1 - \eta - \text{TV}$ reading is the crude corollary. Both feed the same conclusion once the two hypotheses are placed at critical separation.

Theorem 3 (Honest local coverage forces inflation). *Fix $0 < \alpha < 1$ and $0 \leq \eta < \frac{1}{2}$, and let Assumptions 1–3 hold. There are constants $c_1, c_2 > 0$ and n_0 , depending only on the model constants $(\alpha, \eta, K, c_q, g_+, \psi)$, and a fixed constant baseline $\theta_0 \in \mathcal{H}^s(L)$, such that for all $n \geq n_0$ every T_n satisfying honest local coverage (2) over $\mathcal{H}^s(L)$ has, with $r_n := L^{d/(2s+d)}(ng(x_0))^{-s/(2s+d)}$,*

$$\mathbb{P}_{\theta_0}\{T_n(x_0) - q_\alpha(\theta_0(x_0)) \geq c_1 r_n\} \geq c_2 > \frac{1}{2}. \quad (5)$$

In particular the realized threshold exceeds the oracle by order r_n on a majority of samples, so both its median excess and its expected positive excess satisfy

$$\sup_{\theta \in \mathcal{H}^s(L)} S_\theta(T_n) \geq S_{\theta_0}(T_n) \geq c_1 c_2 r_n. \quad (6)$$

Proof. Take the constant baseline $\theta_0 \equiv u_0$ and the bump $\theta_1 = \theta_{h,a}$ of Assumption 3, with $\psi(0) = 1$ so that $q_1 > q_0$. Both lie in $\mathcal{H}^s(L)$.

Indistinguishability. The two laws differ only on the bump support, of covariate measure $g(x_0)h^d(1+o(1))$ by Assumption 1, carrying that many calibration points times n in expectation. Each contributes at most $K\delta^2$ to the Kullback–Leibler divergence by Assumption 2, and the sample is i.i.d., so

$$\text{KL}(P_{\theta_1}^n, P_{\theta_0}^n) \leq ng(x_0)h^d K\delta^2(1+o(1)), \quad \delta = aLh^s.$$

Balance the two scales by choosing $h \asymp (L^2ng(x_0))^{-1/(2s+d)}$, which makes $ng(x_0)h^d\delta^2 = O(1)$ and gives quantile shift

$$q_1 - q_0 \geq c_q \delta = c_q a L h^s \asymp a^{d/(2s+d)} c_q L^{d/(2s+d)} (ng(x_0))^{-s/(2s+d)} = c_1 r_n,$$

the last step using $\delta = aLh^s$ with the chosen h , since $1 - \frac{2s}{2s+d} = \frac{d}{2s+d}$. Fix the amplitude factor $a < a_0$ small enough (a constant, not depending on n) that Pinsker gives $\text{TV}(P_{\theta_1}^n, P_{\theta_0}^n) \leq \tau$

with $\tau < \frac{1}{2} - \eta$, which is possible because the KL bound scales as a^2 . Then Lemma 2 gives $\beta_{1-\eta} \geq 1 - \eta - \tau =: c_2 > \frac{1}{2}$.

Inflation. By the first part of Lemma 2, $\mathbb{P}_{\theta_0}\{T_n(x_0) \geq q_1\} \geq c_2$, and $\{T_n(x_0) \geq q_1\} \subseteq \{T_n(x_0) - q_0 \geq c_1 r_n\}$ because $q_1 - q_0 \geq c_1 r_n$, which is (5). Taking the expected positive part gives $S_{\theta_0}(T_n) \geq c_1 c_2 r_n$, and $\theta_0 \in \mathcal{H}^s(L)$ puts the supremum in (6) at least this high. The threshold n_0 is where the bump support fits inside the neighbourhood of Assumptions 1–3 and $\theta_{h,a}$ stays in the parameter interval. $\square$

The mechanism is the intended one. The data cannot tell a smooth baseline from a small, local increase in residual uncertainty. Exact local protection against the increase forces the set to stay inflated even when the baseline is true, and the size of the forced inflation is the amplitude of the largest undetectable bump. The statement (5) is stronger than an expectation bound: it is the *typical* realized threshold, not an average over rare wide sets, that carries the inflation.

4 Attainability, so the bound is not vacuous

An honest procedure attains the bound, so Theorem 3 is a real optimum. Assumptions 1–3 supply the information bound behind the lower half. The matching upper half needs a family in which the local quantile is estimable at the $m_h^{-1/2}$ rate. In the standard regular models (location, scale, conditional-quantile, and truncated exponential families) it is, by the bias-plus-margin construction. Estimate the local oracle threshold by a boxcar (or kernel) quantile $\hat{q}(x_0)$ over the ball $B(x_0, h)$, holding $m_h \asymp n g(x_0) h^d$ effective calibration points. Under Assumptions 1–3 its bias is at most $C_q L h^s$ (quantile sensitivity applied to the Hölder variation of θ over the ball) and its stochastic fluctuation is $O_P(m_h^{-1/2})$. Set

$$T_n(x_0) = \hat{q}(x_0) + C_q L h^s + \kappa m_h^{-1/2},$$

with $\kappa = \kappa(\eta)$ a fixed multiple of the fluctuation scale. The deterministic bias term covers the systematic error and the margin covers the stochastic error with probability $1 - \eta$, so T_n is honestly locally valid (2), and its excess threshold is $O(L h^s + m_h^{-1/2})$. Minimizing over h gives the modulus (7), so the excess matches (6) up to constants. The honest optimum is therefore exactly the local modulus, achieved by inflating a local quantile estimate by its own bias and margin, and Theorem 3 says no honest procedure does better.

5 The bandwidth reading

The rate is the local modulus of continuity,

$$r_n = L^{d/(2s+d)} (n g(x_0))^{-s/(2s+d)} \asymp \inf_{h>0} \left\{ L h^s + (n g(x_0) h^d)^{-1/2} \right\}, \quad (7)$$

the sum of the largest undetectable local heterogeneity, $L h^s$, and the local sampling scale, $(n g(x_0) h^d)^{-1/2}$, minimized at $h \asymp (L^2 n g(x_0))^{-1/(2s+d)}$. Writing $m_h \asymp n g(x_0) h^d$ for the effective local sample size, the honesty cost is $L h^s + m_h^{-1/2}$: as data thin, both the best neighbourhood and the unavoidable inflation grow. This is the small-sample statement, with no appeal to uniform pooling. Exact local validity does not force wide neighbourhoods; it forces protection against locally undetectable alternatives, whose width is the modulus.

6 Unknown smoothness: honesty cannot adapt

Over a single known class the lower bound (6) is attainable, so coverage and minimax-rate sharpness are not incompatible there. The genuine separation is adaptive.

Theorem 4 (No adaptation under honesty). *Let $0 < s_0 < s_1$ and require honest local coverage over $\mathcal{H}^{s_0}(L) \cup \mathcal{H}^{s_1}(L)$. Then there is $c > 0$ with*

$$\sup_{\theta \in \mathcal{H}^{s_1}(L)} S_\theta(T_n) \geq c (ng(x_0))^{-s_0/(2s_0+d)}, \quad (8)$$

which is strictly slower than the smooth-class rate $(ng(x_0))^{-s_1/(2s_1+d)}$. An honest set cannot achieve the faster sharpness available on the smoother subclass.

Proof. Take the baseline $\theta_0 \equiv u_0$, which is constant and hence lies in $\mathcal{H}^{s_1}(L)$, and the indistinguishable bump alternative from the rougher class $\mathcal{H}^{s_0}(L)$ at $h \asymp (ng(x_0))^{-1/(2s_0+d)}$, $\delta \asymp (ng(x_0))^{-s_0/(2s_0+d)}$. The proof of Theorem 3 applies verbatim and forces $S_{\theta_0}(T_n) \gtrsim \delta \asymp (ng(x_0))^{-s_0/(2s_0+d)}$, even though $\theta_0 \in \mathcal{H}^{s_1}(L)$. $\square$

This is the honesty-adaptation obstruction familiar from nonparametric confidence bands (Low, 1997; Cai and Low, 2004). A sharp point estimator can read apparent smoothness off the data and choose a favourable bandwidth. An honest coverage guarantee cannot trust apparent smoothness, because a rough local alternative is statistically indistinguishable, so it must carry the rougher width. This is the strongest form of the thesis: the local estimator is sharper not by a better estimate but by declining to insure against what it cannot see.

Remark 5 (Average validity is weaker; a heuristic). If validity means only $\inf_\theta \mathbb{E}_\theta F_{\theta(x_0)}(T_n) \geq 1 - \alpha$, averaging over calibration samples, a procedure may compensate a too-narrow set on some samples with a wide set on others, and the first-order argument weakens. The following is a sketch, not a proved bound: it needs the random threshold to stay, with high probability, in a region where $t \mapsto F_u(t)$ is uniformly strongly concave, which we do not verify here. Under that condition Jensen gives $\mathbb{E}_\theta T_n \geq q_\alpha(\theta(x_0))$ and a curvature bound $\text{Var}(T_n) \lesssim \mathbb{E}_\theta T_n - q_\alpha(\theta(x_0))$, so small expected excess would make T_n a low-MSE estimator of the conditional quantile, and the minimax MSE bound then yields $\sup_\theta [\mathbb{E}_\theta T_n - q_\alpha(\theta(x_0))] \gtrsim (ng(x_0))^{-2s/(2s+d)}$. The exponent would double because average coverage is a materially weaker requirement than honest per-sample coverage.

7 An exact check

The bound is a testing statement, so on the Gaussian location model it has no free constants to estimate. Take fixed design $x_i = (i - \frac{1}{2})/n$ and $R_i | X_i \sim N(\theta(x_i), 1)$, baseline $\theta_0 \equiv 0$, and the triangular bump $\psi(z) = (1 - |z|)_+$ with $s = d = 1$. The log-likelihood ratio is exactly Gaussian, so with the chi-square separation $I_n = \sum_i (\theta_1(x_i) - \theta_0(x_i))^2$,

$$\text{TV}(P_{\theta_1}^n, P_{\theta_0}^n) = 2\Phi(\tfrac{1}{2}\sqrt{I_n}) - 1, \quad \beta_{1-\eta}(P_{\theta_1}^n, P_{\theta_0}^n) = \Phi(z_{1-\eta} - \sqrt{I_n}),$$

the second being the exact frontier (4). Along the critical scaling $h = n^{-1/(2s+d)}$, $\delta = h^s$, the separation converges, $I_n \rightarrow \int_{-1}^1 (1 - |z|)^2 dz = \frac{2}{3}$, so both quantities are constant in n : the Le Cam boundary. Below the critical exponent the bump is detectable and $\text{TV} \rightarrow 1$; above it,

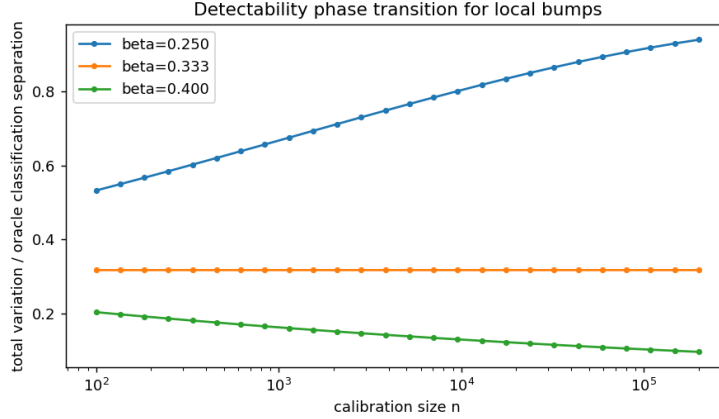


Figure 1: Total variation between the two n -sample laws, for a bump bandwidth $h = n^{-\beta}$. At the critical $\beta = 1/(2s + d) = 1/3$ it is flat at 0.317; below it rises to one (detectable), above it falls to zero (invisible).

invisible and $TV \rightarrow 0$ (Figure 1). At the boundary the forced minimum inflation, as a fraction of the quantile shift $q_1 - q_0 = \delta$, is $\beta_{0.9} = \Phi(z_{0.9} - \sqrt{2/3}) = 0.679$, strictly above the crude Le Cam reading $1 - \eta - TV = 0.583$ and, being above $\frac{1}{2}$, forcing inflation on a majority of samples (Figure 2). The nonadaptation gap of Theorem 4 shows the same way: the forced s_0 -rate over the smooth- s_1 oracle rate diverges as $n^{s_1/(2s_1+d)-s_0/(2s_0+d)} = n^{1/15}$, matched to four figures by the closed form. A random-design exponential experiment reproduces every scaling with Monte Carlo.

8 What it says for conformal prediction

The obstruction is not exchangeability, weighting, or pooling; localized conformal sets that reweight by proximity are marginally valid and can be sharp. The obstruction is indistinguishability under an honesty requirement. Three statements should be kept apart. Zero or $o(r_n)$ oracle excess is incompatible with honesty. Minimax-rate sharpness, $O(r_n)$ excess at a known smoothness s , is compatible with honesty, and Section 4 attains it. The smooth-class rate while honest over a rougher class is again incompatible. Theorem 3 is therefore a nonparametric *inflation* theorem, not a claim that honesty and minimax-rate sharpness never coexist; Theorem 4 is the genuine separation.

The applied reading is the one experience already supplies. At small effective local sample size m_h , the modulus $Lh^s + m_h^{-1/2}$ is large and the honest set is inflated by a correspondingly large amount. When the objective is a sharp predictive score rather than high-probability local coverage, a kernel or local-likelihood estimate of the residual law may be preferable, because it does not carry the one-sided insurance radius honesty requires; whether it wins is a separate decision-theoretic comparison under the chosen score, not a corollary of the width bound. The honest guarantee is not free and it is not distribution-free at a point. Its price is the sharpness it surrenders to insure against the alternatives the data cannot rule out.

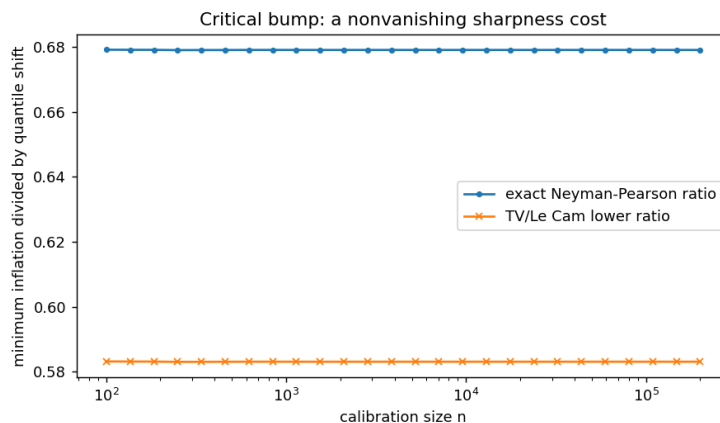


Figure 2: Minimum inflation forced at the smooth baseline, as a fraction of the quantile shift. The exact Neyman–Pearson frontier (0.679) sits well above the crude Le Cam/TV bound (0.583); both are constant in n and above $\frac{1}{2}$.

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